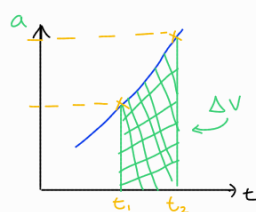
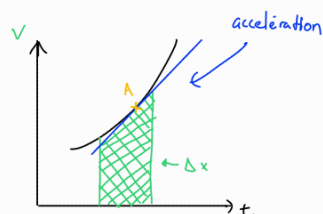
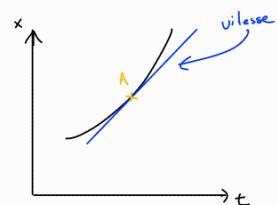


methode graphique



MRU : Vitesse constante $\rightarrow V = c \cdot t$

$$① x_2 = x_1 + v_1(t_2 - t_1)$$

MRUA

$$① x_2 = x_1 + v_1(t_2 - t_1) + \frac{1}{2} a (t_2 - t_1)^2$$

$$② v_2 = v_1 + a(t_2 - t_1)$$

Chute Libre (MRUA)

$$a_j = -g$$

$$① y_2 = y_1 + v_{y1}(t_2 - t_1) - \frac{1}{2} g(t_2 - t_1)^2$$

$$② v_{y2} = v_{y1} - g(t_2 - t_1)$$

Projectile

$$a_x = 0 \quad a_y = -g$$

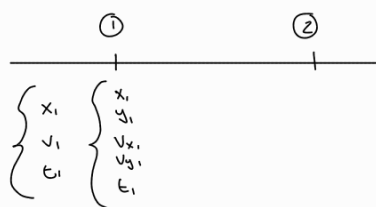
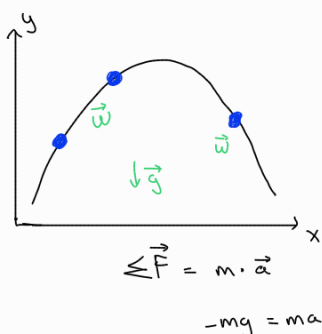
Selon x (MRU)

$$① x_2 = x_1 + v_{x1}(t_2 - t_1)$$

Selon y (MRUA)

$$① y_2 = y_1 + v_{y1}(t_2 - t_1) - \frac{1}{2} g(t_2 - t_1)^2$$

$$② v_{y2} = v_{y1} - g(t_2 - t_1)$$



Exercices

#2

MRUA

$$a = 9.81 \text{ m/s}^2$$

$$c = 3 \cdot 10^8$$

①	②
$x_1 = 0$	$x_2 = ?$
$v_1 = 0$	$v_2 = \frac{1}{10} c$
$t_1 = 0$	$t_2 = ?$

$$① x_2 = 0 + 0(t_2 - 0) + \frac{1}{2} a (t_2 - 0)^2$$

$$② \frac{1}{10} c = 0 + a(t_2 - 0)$$

$$t_2 = 3.06 \cdot 10^6 \text{ s} \rightarrow \text{Jour} \approx 35.42 \text{ jours}$$

$$x_2 = 4.59 \cdot 10^{13} \text{ m}$$

MRUA

$$① x_2 = x_1 + v_1(t_2 - t_1) + \frac{1}{2} a (t_2 - t_1)^2$$

$$② v_2 = v_1 + a(t_2 - t_1)$$

①	MRUA	②
$x_1 = 0$		$x_2 = 150$
$v_1 = 0$		$v_2 = ?$
$t_1 = 0$		$t_2 = ?$

MRUA

$$① x_2 = x_1 + v_1(t_2 - t_1) + \frac{1}{2}a(t_2 - t_1)^2$$

$$② v_2 = v_1 + a(t_2 - t_1)$$

$$① 150 = 0 + 2(t_2 - 0) + \frac{1}{2}a(t_2 - 0)^2$$

$$② v_2 = 2 + a(t_2 - 0)$$

$$t_2 = 12.25 \text{ s}$$

$$v_2 = 24.50 \text{ m/s}$$

$$24.5 \text{ m/s} \quad 88 \text{ km/h} \neq 100 \text{ km/h}$$

Exercice #3 → PHYCHAP4.PDF

①	MRUA	②
$x_1 = 0$		$x_2 = ?$
$v_1 = 13.89 \text{ m/s}$		$v_2 = 0$
$t_1 = 0$		$t_2 = ?$

MRUA

$$① x_2 = x_1 + v_1(t_2 - t_1) + \frac{1}{2}a(t_2 - t_1)^2$$

$$② v_2 = v_1 + a(t_2 - t_1)$$

$$① x_2 = 0 + 13.89(t_2 - 0) + \frac{1}{2}(-10) \cdot (t_2 - 0)^2$$

$$② 0 = 13.89 + -10(t_2 - 0)$$

$$t_2 = 1.39 \text{ s}$$

$$x_2 = 9.65 \text{ m}$$

non vu que pour 50 km/h et
une décélération de 10 m/s^2 la traîne
serais de 9.65 m

#3

3000m		←	
①	②	②	①
$x_1 = 0$	$x_2 = ?$	$x_2 = ?$	$x_1 = 3000$
$v_1 = 27.78$	$v_2 = 0$	$v_2 = 0$	$v_1 = 36.11$
$t_1 = 0$	$t_2 = ?$	$t_2 = ?$	$t_1 = 0$

MRUA

$$① x_2 = x_1 + v_1(t_2 - t_1) + \frac{1}{2}a(t_2 - t_1)^2$$

$$② v_2 = v_1 + a(t_2 - t_1)$$

$$t_2 = 27.78 \text{ s}$$

$$x_2 = 385.86 \text{ m}$$

$$x_2 = 651.97$$

$$t_2 = 36.11$$

$$x_2 - x_1 > 0$$

Pas de collision